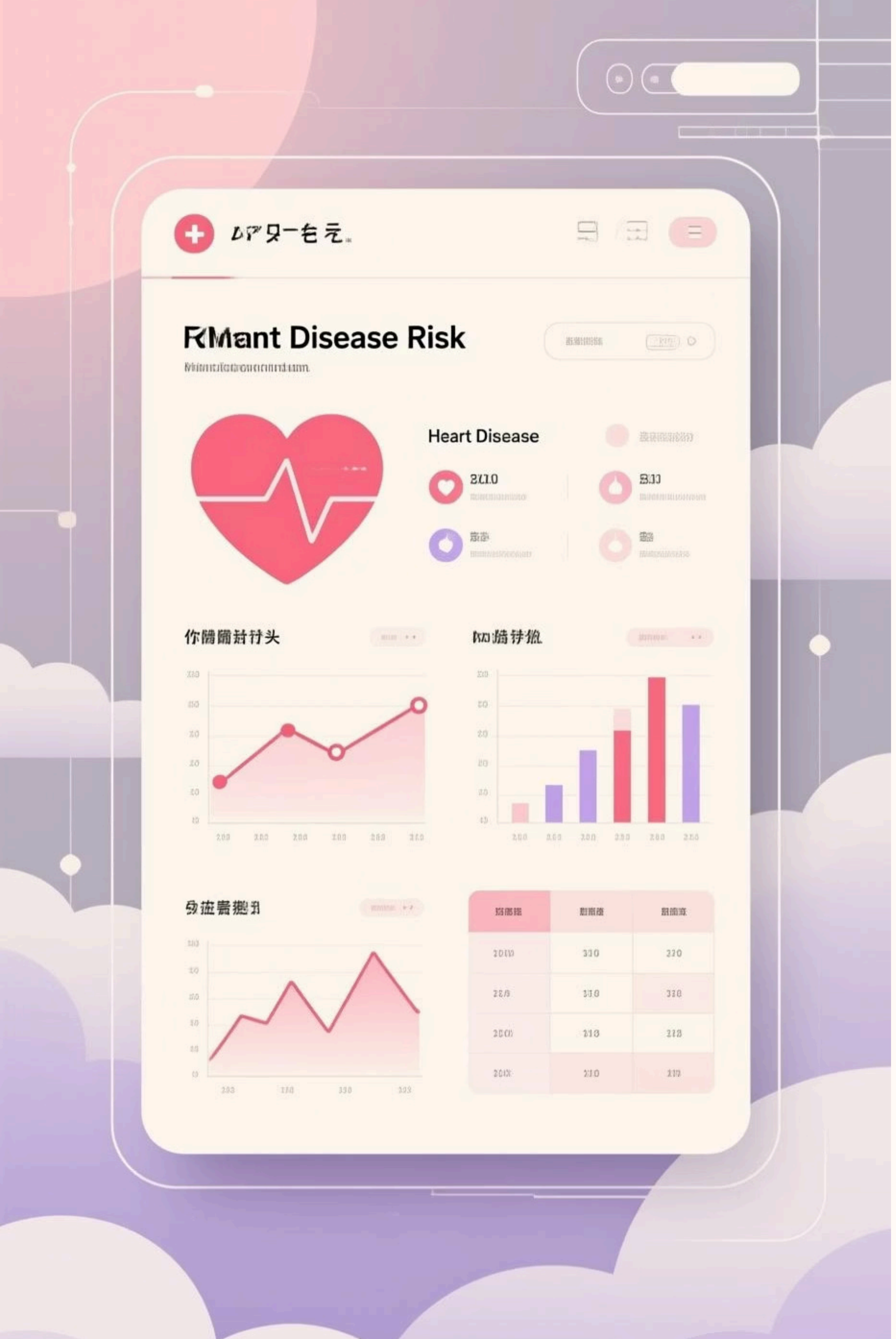


Heart Disease Risk Prediction

A Comparison of Logistic Regression & Random Forest

Problem Statement & ML Objective

Our objective is to accurately predict binary heart disease risk (0/1) in patients, generating a probability score rather than a simple classification.

Quantifying Uncertainty

For clinicians, providing clear risk probabilities.

Stratified Treatment

Facilitating tailored treatment plans based on risk.

Early Intervention

Enabling timely action based on risk thresholds.

Linear Regression: Why it Fails for Binary Classification

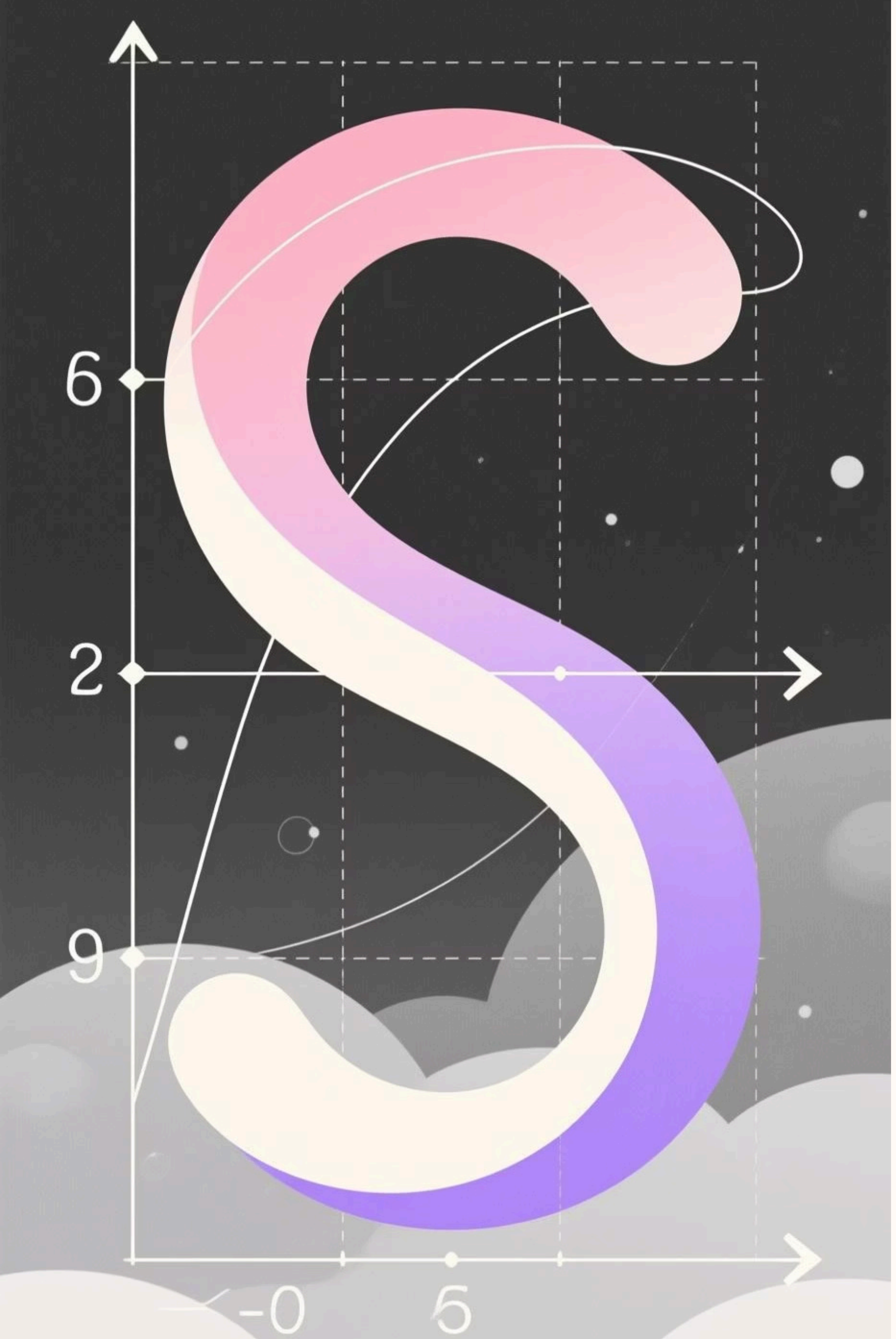
Linear Regression is designed for continuous outcomes, modeling a linear relationship between features and target.

$$\hat{y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n$$

It minimizes Mean Squared Error (MSE) to find optimal coefficients.

$$MSE = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

However, it outputs real values ($-\infty, \infty$), which cannot be directly interpreted as probabilities [0,1], and lacks a natural decision boundary for classification.



Logistic Regression: Core Idea & Sigmoid Transformation

Logistic Regression uses the sigmoid function to transform linear outputs into probabilities [0, 1].

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

This transformation allows for a clear interpretation of risk as a probability, with a threshold (e.g., 0.5) defining the classification decision boundary.

For example, if $z = 0$, $\sigma(0) = 0.5$. If $z = 3$, $\sigma(3) \approx 0.95$.

Logistic Regression: Metrics & Interpretation

1

Confusion Matrix

Summarizes true positives, true negatives, false positives, and false negatives to evaluate model performance.

2

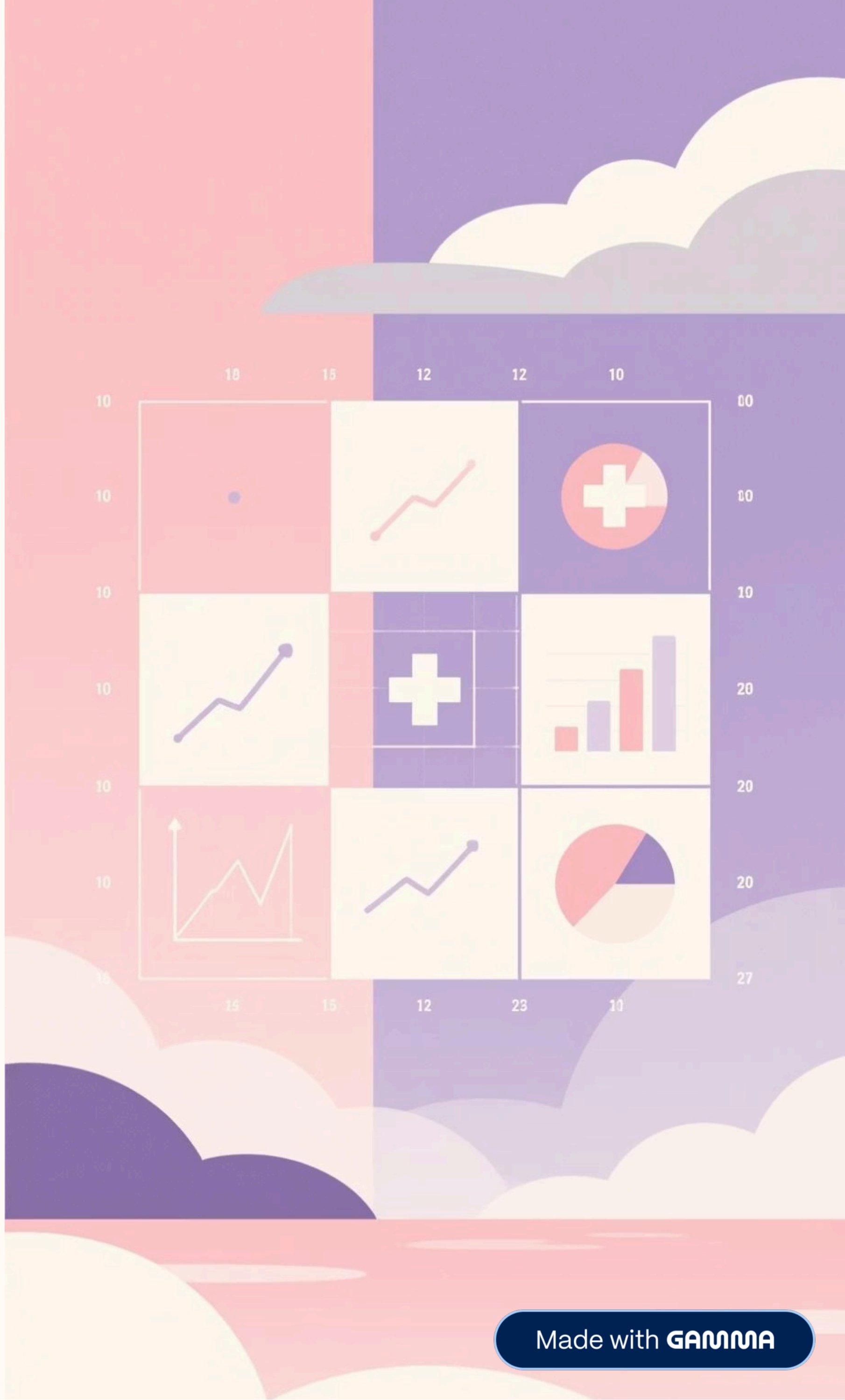
Key Metrics

Precision, Recall, F1-score, and AUC-ROC (Area Under the Receiver Operating Characteristic curve) provide comprehensive performance insights.

3

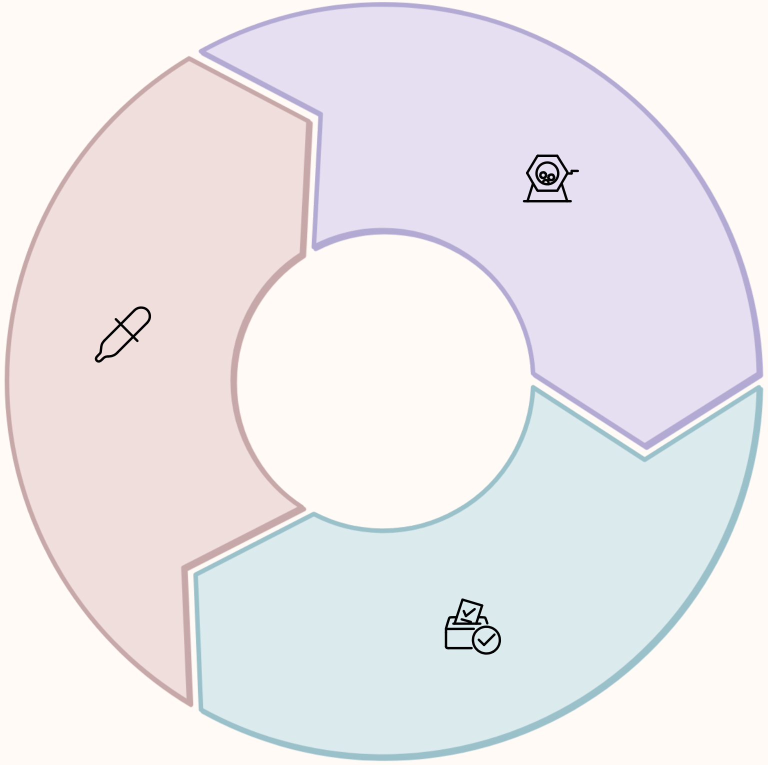
Odds Ratio

Crucial for clinical interpretability, showing the change in odds of heart disease for a one-unit increase in a feature.



Random Forest: Concept & Architecture

Random Forest is an ensemble learning method that builds multiple decision trees for robust classification.



Bagging

Each tree trains on a bootstrap sample (random sampling with replacement).

Feature Randomness

Random subset of features considered at each split, decorrelating trees.

Ensemble Prediction

Final prediction by aggregating individual tree predictions (majority vote).

Random Forest Strengths for Heart Disease Prediction

Non-linearity Handling

Captures complex relationships between medical features and risk.

High Accuracy

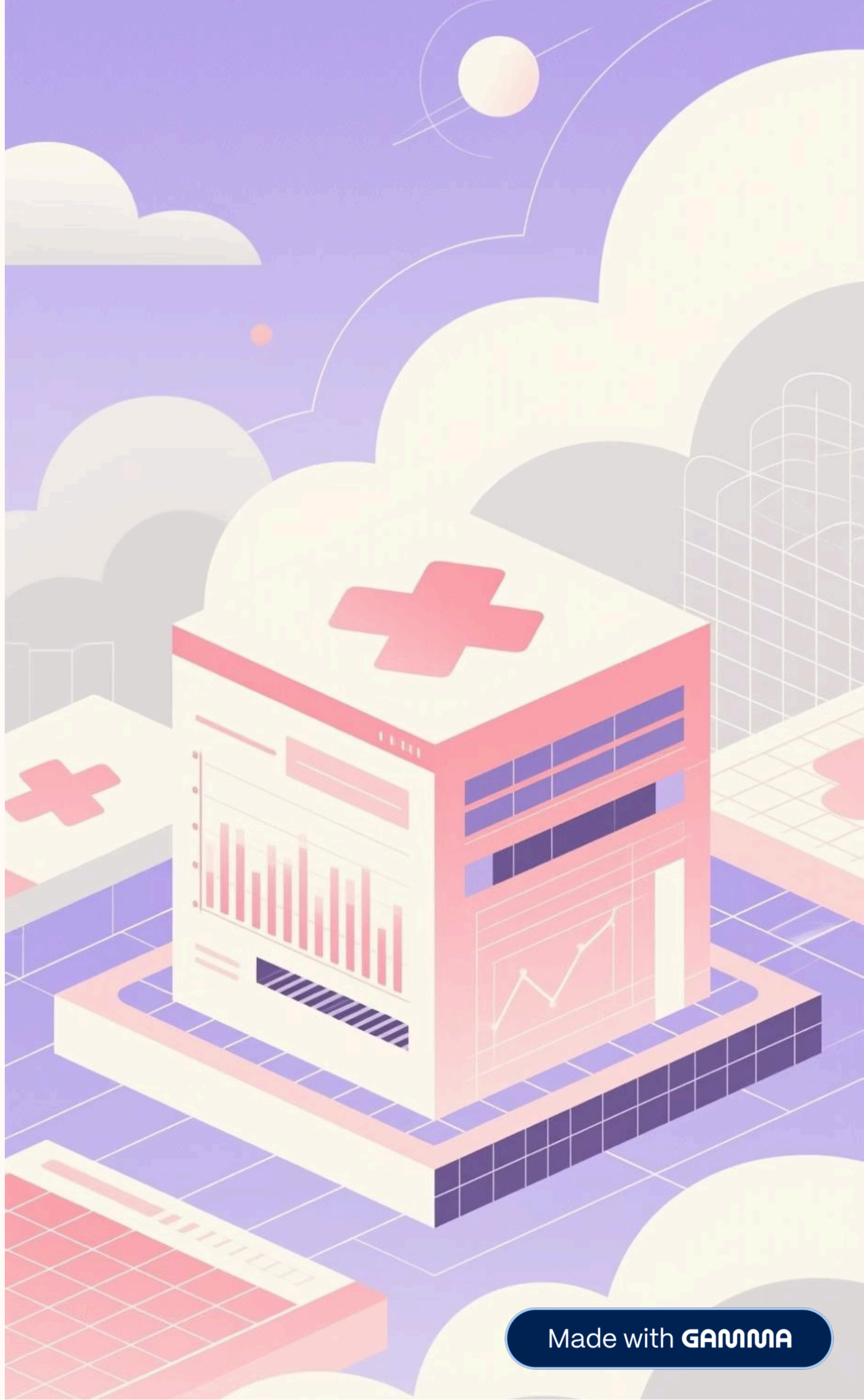
Achieves higher predictive accuracy due to ensemble averaging, reducing overfitting.

Feature Importance

Provides quantifiable scores, highlighting influential physiological parameters.

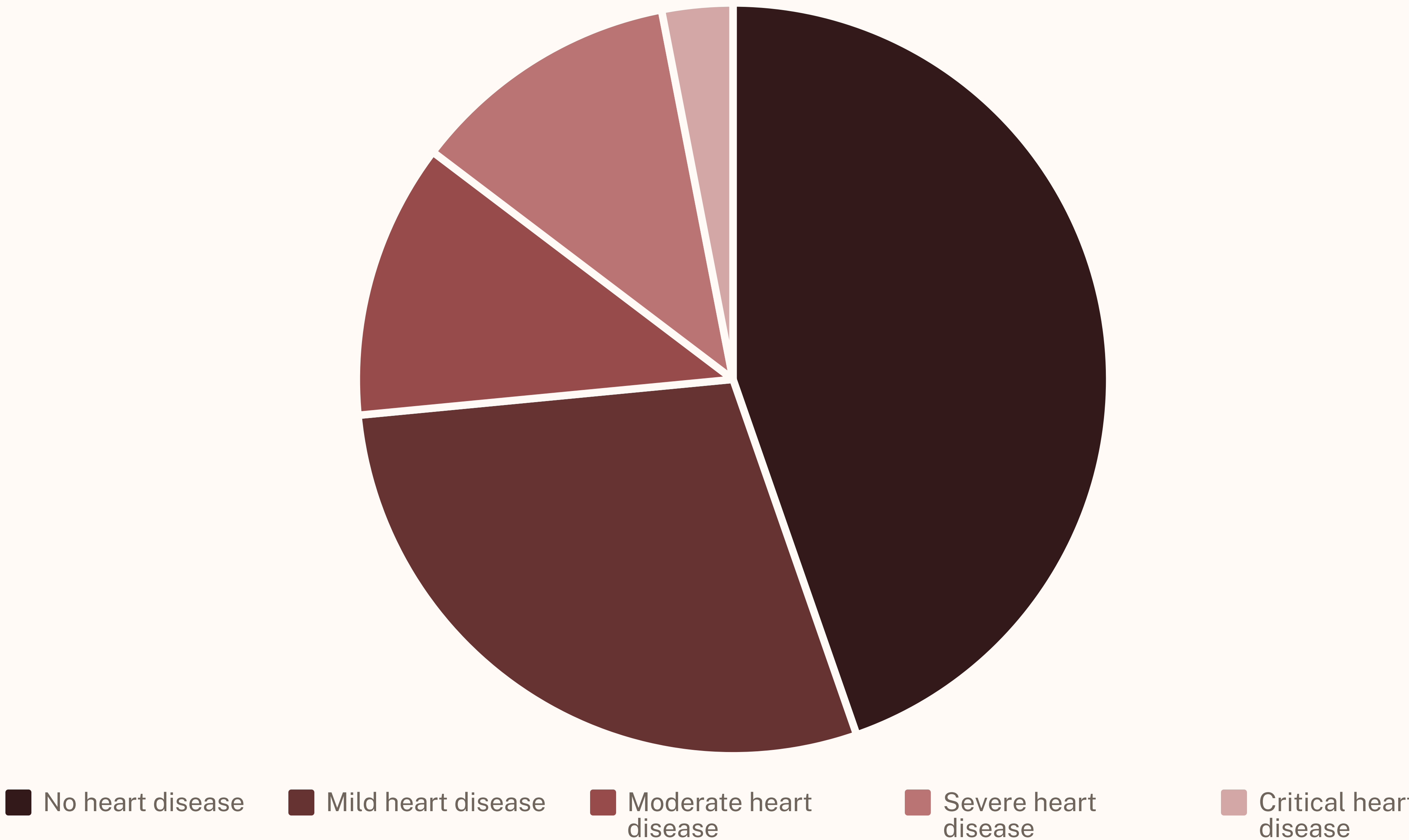
Robustness

Resilient to noise and missing values, suitable for real-world clinical data.



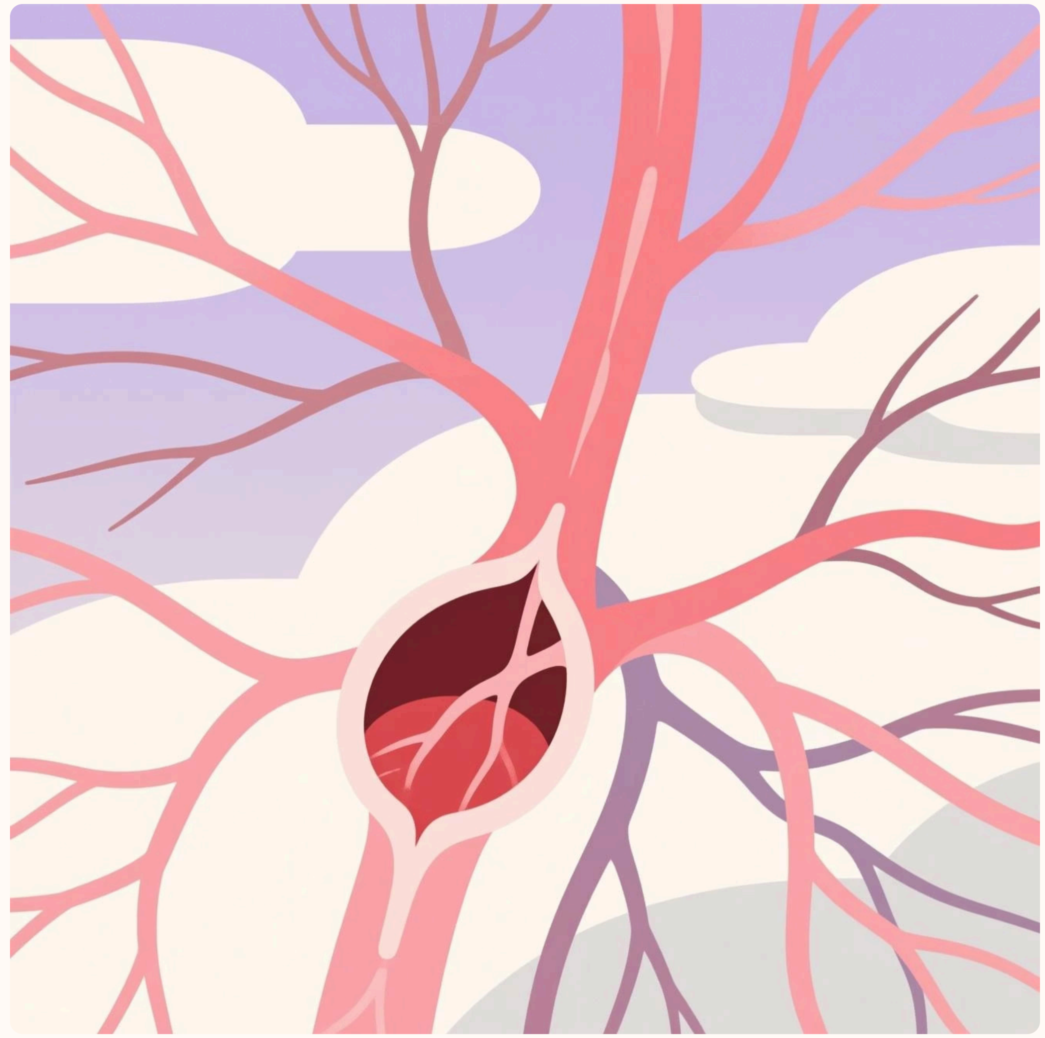
Heart Disease Severity Distribution

The dataset reveals a highly imbalanced distribution of heart disease severity.



Approximately 73.4% of patients have no or mild heart disease, while only about 20% exhibit moderate to critical conditions. Severity increases as the number of patients decreases.

Key Predictors: Oldpeak & Number of Vessels



Oldpeak (ST Depression)

A strong positive linear correlation (0.44) with heart disease severity. Higher ST depression indicates more severe disease.

Also, older individuals tend to experience more ST segment depression (0.26).

Number of Vessels (Ca)

The most important predictor (0.52) of heart disease severity. More blocked vessels directly lead to more severe disease.

Older people are more likely to have increased blockages (0.37).

Both 'oldpeak' and 'ca' are the strongest predictors in this dataset, with 'ca' being the most influential.

Final Model Choice: A Hybrid Approach

Primary: Logistic Regression

- Interpretable coefficients (odds ratios).
- Well-calibrated risk probabilities.
- Transparent and explainable for clinical use.

Secondary: Random Forest

- Validates predictions and highlights discrepancies.
- Captures nonlinearities and complex interactions.
- Higher accuracy for peak predictive performance.

After feature engineering, Logistic Regression achieved 79% accuracy, while Random Forest reached 86% accuracy.